Higgs Fluctuation Solver in Four Dimensions: Part I—A Matrix-Free Global-Scalar Bounce and Determinant-Renormalization Audit

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We report a matrix-free numerical pipeline for semiclassical false-vacuum decay in a four-component global-$O(4)$ scalar theory in four Euclidean dimensions. The calculation combines continuum bounce shooting, relaxation of the sampled profile to a stationary point of the discrete action, channel-resolved fluctuation spectra, deflated stochastic Lanczos quadrature (SLQ), and a continuum partial-wave determinant with Gel'fand–Yaglom radial ratios and large-angular-momentum subtraction. For the benchmark potential, the continuum bounce action is $B = 145.203947$ and the boundary-tail residual is 0.281955%. On an $L = 26$, $h = 0.8$ lattice, the relaxed field has an RMS equation-of-motion residual of 4.43×10^{-12}; its low spectrum contains one negative radial mode, four lattice-lifted translations, and three near-zero internal-orientation modes. The 16-probe bare finite-lattice SLQ trace–log is -2079.9042 ± 6.4504, a regulator-dependent quantity. Under the stated continuum subtraction prescription, the scalar partial-wave calculation returns a primed trace–log of 40.9324 at $\mu = m_F = \sqrt{0.4}$, corresponding to a determinant factor 1.2931×10^{-9}. We show that the leading ρ^8 potential is not closed under its scalar one-loop divergence: local ρ^{10} and ρ^{12} counterterms are required, and their finite renormalized coefficients have not been matched. Thus the reported determinant is a prescription-level result, not a complete matched EFT prediction. Its quoted 0.0300 uncertainty is an estimated hard-coded value rather than an independently established error bar. The translation-normalized scalar semiclassical expression is 5.9974×10^{-70} in lattice energy units to the fourth power; it omits the internal-orientation Jacobian and is not a physical electroweak or gauged decay rate. We summarize tested $SU(2)$ operator, vertex, and pole-algebra infrastructure, while identifying finite full-gauge matching, gauge-sector determinant completion, and controlled continuum/volume limits as open work.

I. INTRODUCTION

Semiclassical false-vacuum decay is governed by a nontrivial Euclidean saddle and by the fluctuation determinant around that saddle. The saddle action controls the leading exponential, whereas the one-loop prefactor requires a careful treatment of the negative mode, translation collective coordinates, internal zero modes, and ultraviolet renormalization [1, 2]. These ingredients are individually well understood in analytic and partial-wave calculations, but their simultaneous numerical treatment becomes demanding when the fluctuation operator is large, the background is spatially varying, and lattice and continuum regulators must be related.

This work documents a computational pipeline developed for a four-component real scalar field with a global $O(4)$ symmetry. The production result is a scalar-theory benchmark, not a calculation of the Standard Model Higgs decay rate. The solver combines an $O(4)$-symmetric continuum bounce with a four-dimensional periodic lattice representation, matrix-free Hessian operators, channel-by-channel low-mode extraction, a stochastic trace–log, and a continuum-subtracted radial determinant. The purpose of Part I is to describe this scalar benchmark and the verified computational components without conflating finite-lattice diagnostics with a physical decay prefactor.

Recent work has advanced several neighboring parts of Higgs-vacuum decay. Perturbative studies have treated the Standard Model bounce with running couplings and gravity, computed complete one-loop rates in scalar quartic–quartic models, analyzed gauge–Goldstone mixing and gradient effects in gauged scalar theories, developed non-thin-wall one-loop corrections for scalar bounces, and derived four-dimensional determinant-renormalization formulas for fields of spin 0, 1/2, and 1 [3–7]. A recent Standard Model calculation has also revisited the full one-loop gauge prefactor and transverse-mode counting [8]. Lattice-based Higgs-vacuum studies have examined real-time stability during post-inflationary kination; a distinct Euclidean-lattice Monte Carlo line has proposed decay estimators demonstrated in metastable quantum-mechanical systems [9–11]. The present calculation is complementary to these efforts: it combines a continuum $O(4)$ bounce and continuum-subtracted radial determinant with a relaxed finite-lattice background, matrix-free Hessian analysis, and a bare SLQ trace–log diagnostic. It does not include the Standard Model running-potential and gravitational effects, constitute a Euclidean field-theory Monte Carlo decay measurement, or provide a fully matched gauge-sector determinant. The scalar result is a scaled global-$O(4)$ benchmark, with the finite ρ^{10} and ρ^{12} coefficients still unmatched, rather than a precision electroweak decay rate.

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The project also contains a growing quadratic and perturbative $SU(2)$ extension. Several operator identities, homogeneous limits, vertex generators, and ultraviolet pole relations have been tested. The scalar one-loop analysis additionally shows that the tree-level potential truncated at ρ^8 is not closed under the retained fluctuation determinant: ρ^{10} and ρ^{12} counterterms are required, and their finite coefficients have not been matched. Selected finite-matching pilots have been evaluated, but the finite gauge-sector coefficient set and a complete gauge–Goldstone–ghost determinant are not available. We therefore treat the determinant and matching results as implementation status and prerequisite checks, not as a fully matched EFT or gauged prediction. A later manuscript(Part II) would be justified only after those missing calculations and their systematic checks are completed.

The calculation's main stages and the distinct roles of its lattice and continuum determinant analyses are summarized in Fig. 1. The two determinant routes are complementary rather than sequential: the bare lattice trace–log is a regulator-dependent diagnostic, whereas the continuum partial-wave calculation supplies the reported subtraction- prescription result.

II. MODEL AND SEMICLASSICAL OBSERVABLE

We consider a real four-component field $\mathbf{\Phi}$ with radial amplitude $\rho = (\mathbf{\Phi}\cdot\mathbf{\Phi})^{1/2}$ and Euclidean action

$$S_E[\mathbf{\Phi}] = \int \mathrm{d}^4x \left[\frac{1}{2}\partial_\mu\mathbf{\Phi}\cdot\partial_\mu\mathbf{\Phi} + V(\rho)\right]. \tag{1}$$

The benchmark potential is

$$V(\rho) = -\frac{\mu^2}{2}\rho^2 + \frac{\lambda}{4}\rho^4 + \frac{\gamma}{6}\rho^6 + \frac{\delta}{8}\rho^8, \tag{2}$$

with

$$\mu^2 = -0.40, \qquad \lambda = -2.00, \qquad \gamma = 0.80, \qquad \delta = 0.08. \tag{3}$$

The resulting false-vacuum minimum, barrier, and deeper minimum are shown in Fig. 2. The false vacuum is at $\rho = 0$, with $m_F^2 = V''(0) = -\mu^2 = 0.4$. The negative quartic produces a barrier, while the higher powers stabilize the potential at large field amplitude. All benchmark values are quoted in the solver's scaled units.

Under the usual assumptions for scalar false-vacuum decay, the least-action solution is $O(4)$ symmetric [12] and obeys

$$\rho_B''(r) + \frac{3}{r}\rho_B'(r) = V'(\rho_B), \rho_B'(0) = 0, \qquad \rho_B(\infty) = 0. \tag{4}$$

Its action relative to the false vacuum is

$$B = 2\pi^2 \int_0^\infty r^3 \left[\frac{1}{2}\rho_B'(r)^2 + V(\rho_B(r)) - V(0)\right] \mathrm{d}r. \tag{5}$$

For a scalar theory, the semiclassical expression has the schematic form

$$\frac{\Gamma}{V_4} = \left(\frac{B}{2\pi}\right)^2 \left|\frac{\det{}' H_B}{\det H_F}\right|^{-1/2} e^{-B}\, \mathcal{J}_{\text{internal}}\,[1+O(\hbar)], \tag{6}$$

where H_B and H_F are the bounce and false-vacuum Hessians. The prime denotes removal of collective-coordinate eigenvalues, and the unique negative mode is retained through the absolute determinant. Equation (6) is included to define the target observable; the calculation reported here does not determine $\mathcal{J}_{\text{internal}}$ or a physical energy conversion for the rate.

III. NUMERICAL METHODS

A. Continuum bounce

The bounce equation is integrated from a small positive ϵ using the regular origin expansion

$$\rho_B(\epsilon) = \rho_0 + \frac{V'(\rho_0)}{8}\epsilon^2, \qquad \rho_B'(\epsilon) = \frac{V'(\rho_0)}{4}\epsilon. \tag{7}$$

An overshoot/undershoot bisection in ρ_0 isolates the separatrix approaching the false vacuum. Integration terminates when the field enters a specified physical tail; this avoids extending an approximate separatrix beyond the range where numerical roundoff remains controlled. The action is then evaluated from Eq. (5) using the S^3 angular measure $2\pi^2 r^3$.

B. Periodic lattice and matrix-free Hessian

The lattice uses periodic boundary conditions and a nearest-neighbor four-dimensional Laplacian,

$$(-\Delta_h f)_x = \frac{1}{h^2}\left[8f_x - \sum_{\mu=1}^{4}(f_{x+\hat{\mu}} + f_{x-\hat{\mu}})\right]. \tag{8}$$

The stencil is applied directly to reshaped arrays; the global Hessian is exposed as a matrix-free linear operator rather than assembled. At $L = 26$, one scalar channel has $L^4 = 456{,}976$ sites and the four-component field has 1,827,904 degrees of freedom. The sampled continuum profile is relaxed against the discrete action using a Newton–MINRES method. The false-vacuum operator is diagonalized analytically in the periodic Fourier basis, which also provides a reference for checking the lattice indexing and degeneracies.

For a radial background, the Hessian splits into one longitudinal and three degenerate transverse channels,

$$H_L = -\Delta_h + U_L(x), \quad U_L = -\mu^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \tag{9}$$

$$H_T = -\Delta_h + U_T(x), \quad U_T = -\mu^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6. \tag{10}$$

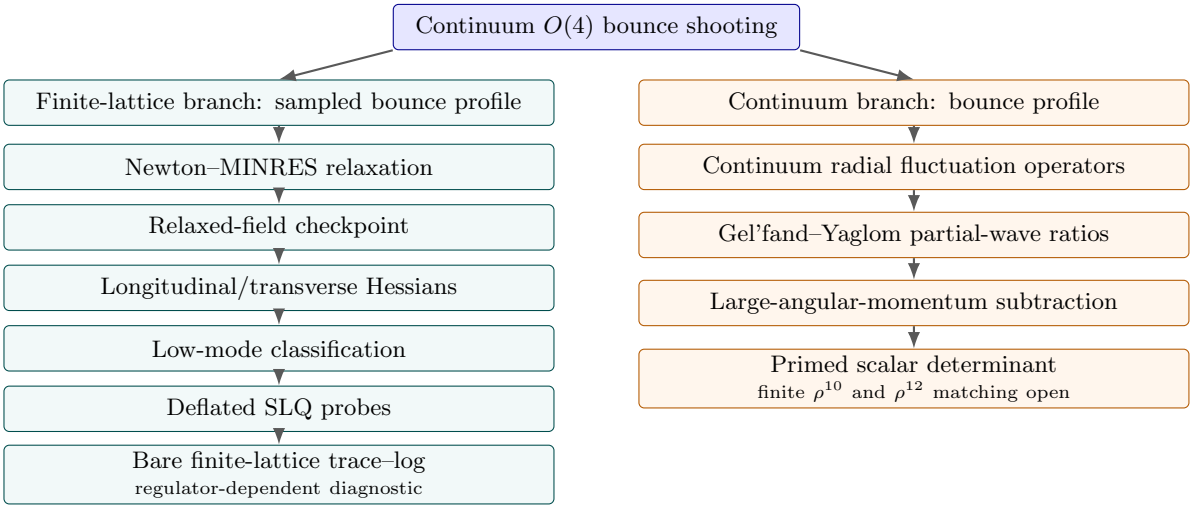


FIG. 1. Overview of the scalar calculation. The continuum bounce seeds the finite-lattice relaxation and is also the input to the continuum radial calculation. The finite-lattice SLQ trace–log is a regulator-dependent diagnostic; the partial-wave route gives the quoted continuum-subtracted determinant under the stated prescription. These are complementary routes, not successive steps, and the determinant is not a complete matched EFT prediction because the finite ρ^{10} and ρ^{12} counterterms remain unmatched.

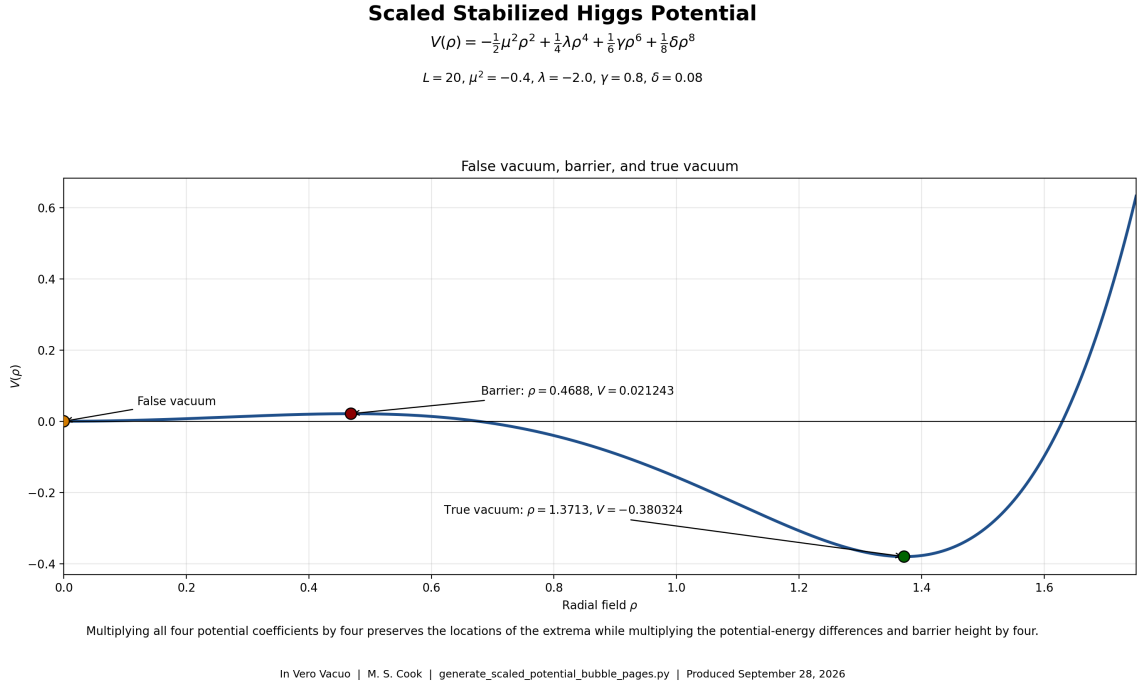


FIG. 2. Benchmark radial potential in the solver's scaled units. The plot marks the false vacuum at $\rho = 0$, the barrier, and the deeper nonzero minimum for the coefficients in Eq. (3). This is the potential-shape benchmark used to define the continuum bounce problem.

The transverse degeneracy is applied analytically, while eigenpairs are computed separately in each channel. This avoids cutting through a degenerate multiplet during low-mode extraction.

C. Lattice-to-physical scaling and run modes

The solver has two distinct spacing conventions. Exploratory research mode uses an explicit, dimensionful

lattice spacing a, with conversions

$$\begin{aligned} \Phi_{\rm phys} &= \frac{\Phi_{\rm lat}}{a}, \\ m^2_{\rm phys} &= \frac{m^2_{\rm lat}}{a^2}, \\ V_{4,\rm phys} &= (La)^4, \\ \Lambda_{\rm UV} &\sim \frac{\pi}{a}. \end{aligned} \tag{11}$$

By contrast, the scalar production results reported here—including bounce relaxation, Hessian spectra, and SLQ—use dimensionless solver units and a refinement spacing h; the main benchmark has $h = 0.8$. The spacing campaign varies h while holding the box extent Lh fixed in these scaled units. Thus h is not the dimensionful spacing a in Eq. (11), and no substitution $a = h$ is implied. The continuum partial-wave calculation uses m_F as its reference scale in the solver's scaled convention. Consequently, the scalar prefactor remains in lattice-energy units to the fourth power; conversion to a physical rate in GeV^4 requires an explicit physical scale and is not part of the reported production calculation.

D. One-loop closure of the radial potential

The tree-level benchmark in Eq. (2) is truncated at ρ^8, but that truncation is not closed under the scalar one-loop divergence when the ρ^6 and ρ^8 interactions are retained nonperturbatively in the leading fluctuation operators. For the radial channels in Eq. (10), the local scalar divergence is proportional to $U_L^2 + 3U_T^2$. Writing $m_F^2 = -\mu^2$, its field-dependent polynomial is

$$\begin{aligned} U_L^2 + 3U_T^2 = {} & 4m_F^4 + 12\lambda m_F^2\rho^2 + (12\lambda^2 + 16\gamma m_F^2)\rho^4 \\ & + (36\lambda\gamma + 20\delta m_F^2)\rho^6 + (28\gamma^2 + 48\lambda\delta)\rho^8 \\ & + 76\gamma\delta\rho^{10} + 52\delta^2\rho^{12}. \end{aligned} \tag{12}$$

The field-independent term cancels in the bounce-minus-false-vacuum action. The final two terms do not belong to the original potential, so a renormalization-closed scalar basis at this loop order must also allow

$$V_{\rm EFT}(\rho;\mu_R) = V(\rho;\mu_R) + \frac{c_{10}(\mu_R)}{10}\rho^{10} + \frac{c_{12}(\mu_R)}{12}\rho^{12}. \tag{13}$$

This is an order-by-order closure statement: c_{10} and c_{12} are one-loop-order counterterm coefficients and, in the adopted counting, do not enter the leading bounce equation or Hessian. Their finite renormalized values nevertheless require matching conditions and have not been fixed in the benchmark reported here. Promoting these operators to independent leading interactions would require a still larger basis at higher orders: for example, the curvature of a leading ρ^{12} term scales as ρ^{10}, whose square generates a ρ^{20} contribution. This scalar result is not a closure proof for the full gauged theory, which also has derivative and gauge-sector counterterms. The need to include loop-generated local operators in the EFT basis is standard in order-by-order effective-theory renormalization [19, 22].

E. Low modes and stochastic trace–log

For a positive operator, $\log\det H = \mathrm{Tr}\log H$. On the bounce, the negative and collective-coordinate modes require explicit classification; the low eigenpairs are therefore computed and deflated before estimating the remaining trace. For independent Rademacher probes z_k, the trace is estimated by

$$\mathrm{Tr}\, f(H) \simeq \frac{1}{N_p}\sum_{k=1}^{N_p} z_k^T f(H) z_k, \qquad f(H) = \log H. \tag{14}$$

The probe average is a Hutchinson-type stochastic trace estimator; Lanczos iteration maps each quadratic form to a Gaussian quadrature on a small tridiagonal matrix [20, 21]. The same probe stream is used for the bounce and false-vacuum operators so that their difference benefits from correlated stochastic fluctuations. Exact low modes are restored analytically after projection.

F. Continuum partial-wave determinant

For an $O(4)$ background, the radial operators are organized by angular momentum ℓ with degeneracy $g_\ell = (\ell+1)^2$. Radial Gel'fand–Yaglom ratios replace functional determinants by ratios of solutions of initial value problems [13, 18]; the large-ℓ asymptotics are subtracted before summation [16]. Translation and orientation zero modes are handled by a positive spectral shift, division by the known powers of that shift, and extrapolation toward zero, following standard primed-determinant treatments [14, 15, 17].

IV. SCALAR BENCHMARK RESULTS

A. Bounce and discrete stationary field

For Eqs. (2)–(3), the continuum shooting solution has center value $\rho_B(0) = 1.367393286$ and action $B = 145.2039474492$. The reported finite-box boundary residual is 0.281955%, below the adopted one-percent criterion. The leading tunneling factor is $e^{-B} = 8.6841 \times 10^{-64}$. The radial bounce and radial action density for this continuum solution are shown in Fig. 3.

The lattice calculation uses $L = 26$ and refinement spacing $h = 0.8$. For the regenerated relaxed field, the RMS and maximum discrete equation-of-motion residuals are 4.43266×10^{-12} and 6.30906×10^{-11}, respectively. The identity relating the transverse Hessian to

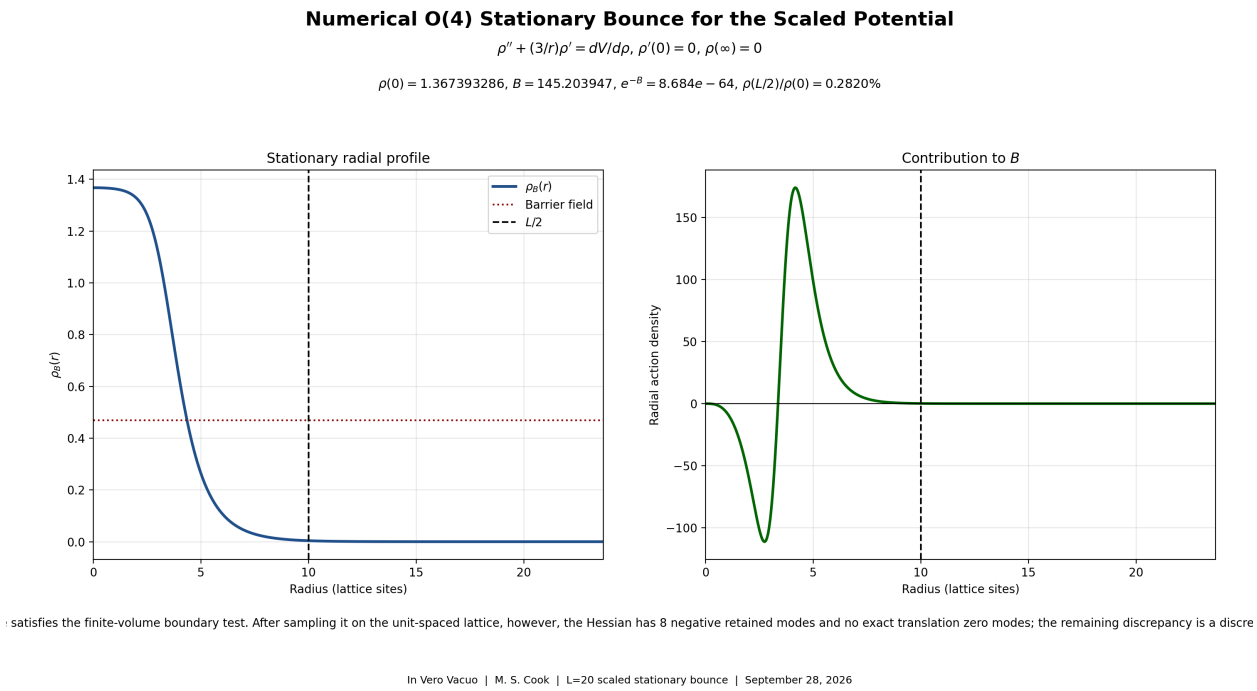


FIG. 3. Continuum $O(4)$ bounce profile and radial contribution to the action for the benchmark potential. The profile begins at $\rho_B(0) = 1.367393286$ and approaches the false vacuum in the tail; the dashed vertical line marks the half-box radius used in the $L = 20$ boundary test. The reported boundary-tail residual is 0.2820%.

the residual provides an additional check of the internal-orientation zero mode. The relaxation history and the relaxed field along a lattice axis are shown in Fig. 4.

B. Collective-coordinate spectrum

The low spectrum has the expected pattern: one negative longitudinal mode, four translation modes lifted slightly below zero by the finite lattice, and three internal-orientation modes close to zero. The reported eigenvalues are summarized in Table I. The translation eigenvalues are not exact zeros because continuous translation symmetry is broken by the fixed lattice. The orientation triplet is protected by the global symmetry and is numerically zero at the relaxation tolerance. The channel-resolved low eigenvalues are visualized in Fig. 5.

C. Bare SLQ and continuum-subtracted determinant

The production SLQ calculation used 16 Rademacher probes, 12 deflated eigenpairs per bounce channel, and Lanczos orders 20, 40, and 60. The order-60 bare finite-lattice trace–log was

$$\log \left| \frac{\det' H_B}{\det H_F} \right|_{\text{bare}} = -2079.9042 \pm 6.4504. \tag{15}$$

TABLE I. Principal scalar benchmark results. Values are in the solver's scaled units unless stated otherwise. The continuum determinant is a prescription-level result because the finite c_{10} and c_{12} matching coefficients are not fixed. Its 0.0300 uncertainty is the source's estimate, not an independently calibrated error.

Quantity	Result
Continuum center field	1.367393286
Continuum bounce action	145.2039474492
Boundary residual	0.281955%
Discrete RMS / max residual	4.43×10^{-12} / 6.31×10^{-11}
Negative longitudinal mode	−0.1850415
Four translation modes	−0.00212653 each
Three orientation modes	-5.33×10^{-12} each
Bare finite-lattice SLQ trace–log	-2079.9042 ± 6.4504
Continuum-subtracted scalar trace–log	40.9324 (estimated ±0.0300)
Continuum-subtracted determinant factor	1.2931×10^{-9}

The quoted spread is the reported finite-lattice estimator uncertainty. This quantity is dominated by ultraviolet regulator dependence and is not a physical enhancement of the decay rate. It cannot be compared directly with the continuum-renormalized result without matching the regulator and counterterms. Figure 6 shows the bare estimate as a function of Lanczos order; its error bars

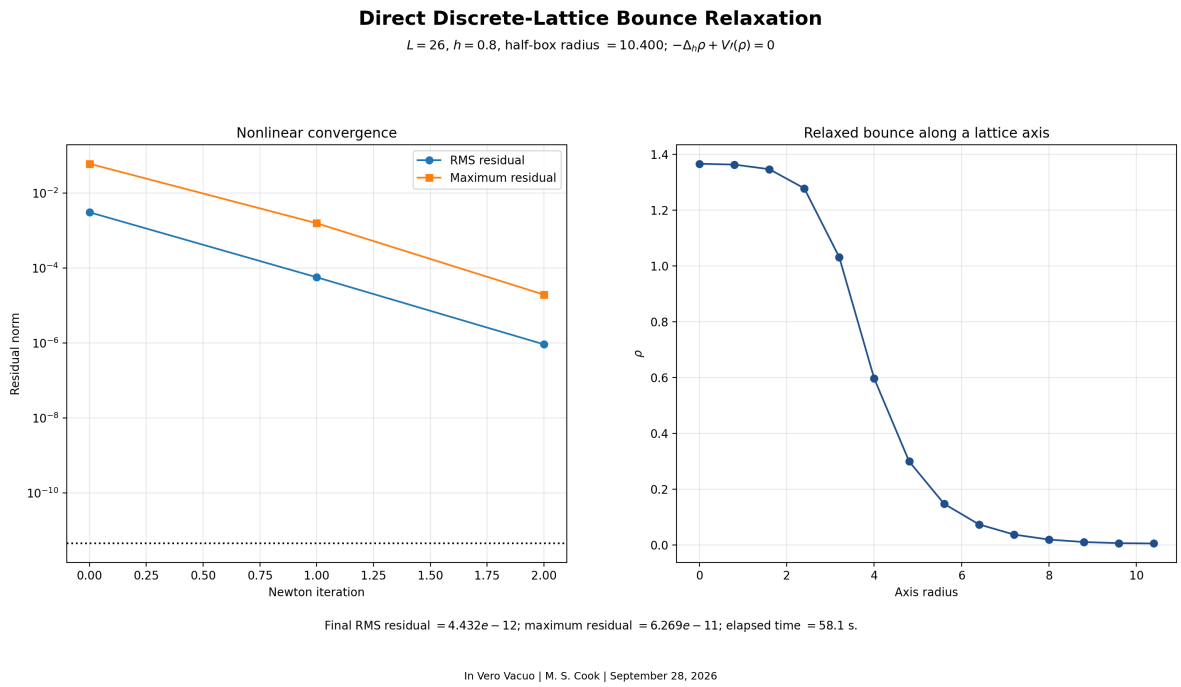


FIG. 4. Direct relaxation of the sampled bounce on the periodic $L = 26$, $h = 0.8$ lattice. The left panel shows the decrease of the RMS and maximum equation-of-motion residuals during Newton–MINRES iterations; the right panel shows the converged radial field along a lattice axis. The final RMS residual is approximately 4.43×10^{-12}.

emphasize the stochastic spread at the probe count used here.

The separate continuum calculation uses a radial cutoff of 20, $\ell_{\max} = 70$, and renormalization scale $\mu = m_F = \sqrt{0.4}$. It reports

$$\log \left| \frac{\det{}' H_B}{\det H_F} \right|_{\overline{\mathrm{MS}}} = 40.9324, \tag{16}$$

$$\left| \frac{\det{}' H_B}{\det H_F} \right|_{\overline{\mathrm{MS}}}^{-1/2} = 1.2931 \times 10^{-9}. \tag{17}$$

The source records an uncertainty estimate of 0.0300 for the trace–log, but that value is hard-coded and is not the output of a complete systematic error analysis. Accordingly, the central value is a benchmark result of the implemented prescription; its precision should not be overstated. There is also a separate renormalization-closure limitation: with nonzero γ and δ, Eq. (12) contains local ρ^{10} and ρ^{12} terms. The determinant's asymptotic subtraction produces a finite number for the specified operator, but the finite renormalized coefficients $c_{10}(\mu_R)$ and $c_{12}(\mu_R)$ have not been fixed by matching. Thus 40.9324 is a prescription-level determinant result for the chosen leading operator, not a complete matched one-loop EFT prediction. The partial-wave sum and positive-shift zero-mode extrapolation are shown in Fig. 7.

Including the continuum bounce action and the four-translation factor gives the scalar partial expression

$$\left(\frac{B}{2\pi}\right)^2 \left| \frac{\det{}' H_B}{\det H_F} \right|_{\overline{\mathrm{MS}}}^{-1/2} e^{-B} = 5.9974 \times 10^{-70}, \tag{18}$$

$$\text{(lattice energy units)}^4. \tag{19}$$

Equation (19) omits the internal-orientation group-volume Jacobian and does not include an explicit conversion to a physical energy scale. It is therefore not a complete scalar decay rate, and it is not a gauged-Higgs prediction. It also inherits the unresolved finite c_{10} and c_{12} matching in the one-loop scalar potential.

V. STATUS OF THE QUADRATIC AND PERTURBATIVE $SU(2)$ EXTENSION

The solver includes a research branch for quadratic gauge–Goldstone and ghost operators around a radial Higgs background with zero background gauge field. The gauge operator is derived from lattice links rather than from a centered continuum derivative; positive background-R_ξ gauges are supported. Homogeneous Fourier spectra, normalization conventions, and Ward-identity residuals have been tested at $\xi = 0.5, 1, 2$. In a regenerated $L = 26$ stationary background, the largest reported on-shell Ward residual was 5.52×10^{-14}. The homogeneous regulated determinant test also approaches

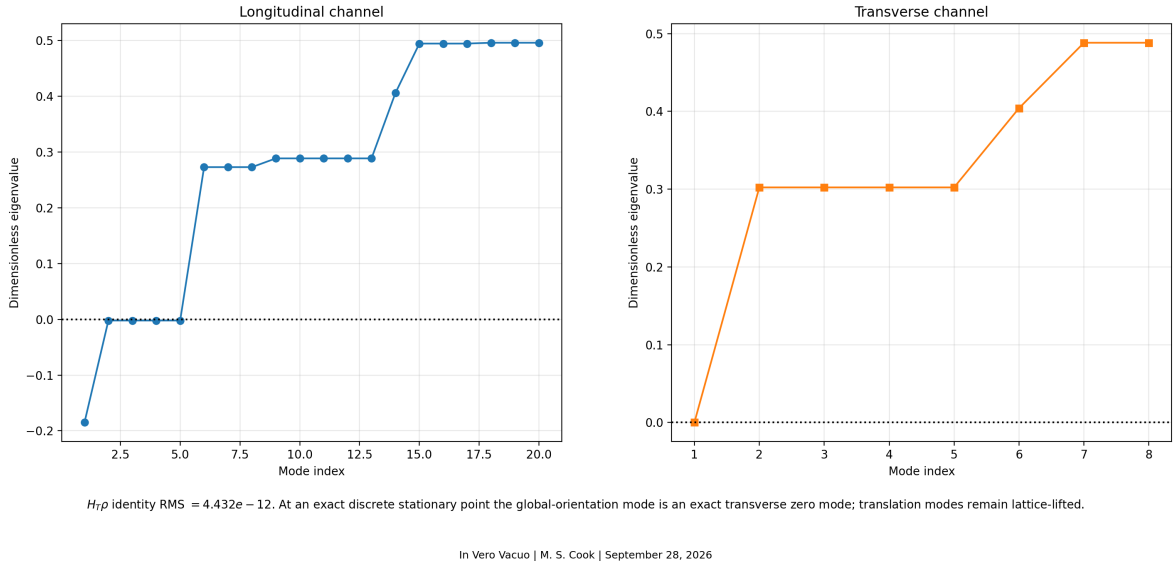


FIG. 5. Low eigenvalues of the relaxed discrete-bounce Hessian in the longitudinal and transverse channels. The longitudinal channel contains one negative mode and four near-zero translation modes; the transverse channel contains the global-orientation zero mode, with its threefold internal degeneracy. The dotted lines indicate zero eigenvalue.

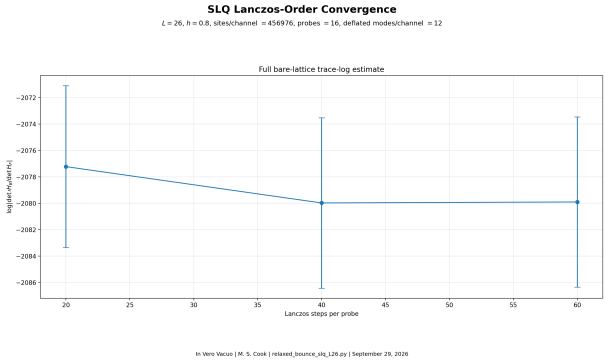


FIG. 6. Lanczos-order dependence of the bare finite-lattice trace–log estimate for $L = 26$, $h = 0.8$, 16 Rademacher probes, and 12 deflated eigenpairs per channel. Error bars show the reported estimator uncertainty. The plotted quantity remains regulator dependent and is not the continuum-renormalized determinant.

gauge-parameter independence as its auxiliary IR shift is removed.

A. Wilson plaquette action and vertex convention

The nonlinear Wilson-gauge vertices cited below use the bare-coupling convention g_0 and generators $T^a = \sigma^a/2$. The link and oriented plaquette variables are

$$U_{x\mu} = \exp[-iag_0 A^a_{x\mu} T^a], \tag{20}$$

$$U_{x,\mu\nu} = U_{x\mu} U_{x+a\hat{\mu},\nu} U^\dagger_{x+a\hat{\nu},\mu} U^\dagger_{x\nu}. \tag{21}$$

The Wilson action is

$$S_g = \frac{4}{g_0^2} \sum_x \sum_{\mu<\nu} \left(1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr} U_{x,\mu\nu}\right), \tag{22}$$

which has continuum normalization $\frac{1}{4}\int \mathrm{d}^4x\, F^a_{\mu\nu} F^a_{\mu\nu}$ for this generator convention [23]. Figure 8 depicts the ordered link product. This action specifies the perturbative vertex-generator convention; it does not imply that a full gauge-sector determinant or physical gauged decay prefactor has been computed.

B. Contained-background gauge–Goldstone diagnostic

A separate diagnostic run dated September 30, 2026, tested the coupled gauge–Goldstone and ghost operators on a contained Gaussian profile. This profile is not a stationary Coleman bounce, and the run is distinct from the $L = 26$ on-shell Ward-identity test above; its spectrum is an operator check, not a determinant ratio or a decay-rate calculation.

In background 't Hooft–Feynman gauge, $\xi = 1$, the matrix-free block for each of the three identical $SU(2)$

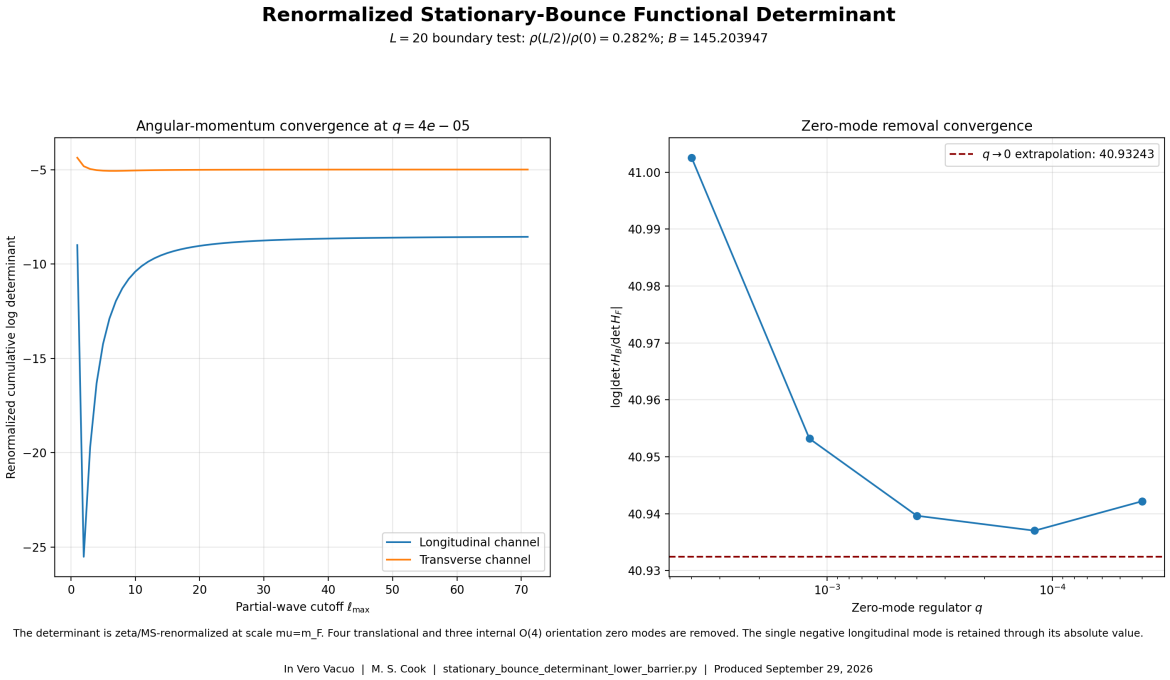


FIG. 7. Continuum determinant convergence diagnostics for the $L = 20$ stationary-bounce benchmark. The left panel shows the asymptotically subtracted partial-wave sum as the angular-momentum cutoff increases; the right panel shows the positive-regulator extrapolation used to remove the four translation and three internal-orientation zero modes. The plotted $q \to 0$ intercept is approximately 40.9324. The source's quoted 0.0300 uncertainty is an estimated, hard-coded value, not an uncertainty inferred from this figure.

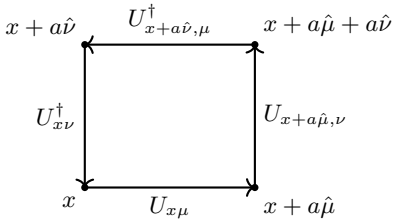


FIG. 8. One oriented Wilson plaquette. The arrows follow the ordered product in Eq. (21); daggered links traverse opposite to their positive coordinate directions. Its contribution to the action is weighted by the corresponding term in Eq. (22).

generators acts on $(A_1, A_2, A_3, A_4, \chi)$:

$$\mathcal{H}_{A\chi} = \begin{pmatrix} (-\Delta + m_W^2)\delta_{\mu\nu} & 2q\,\partial_\mu\rho \\ 2q\,\partial_\nu\rho & -\Delta + V_T'' + m_W^2 \end{pmatrix}. \qquad (23)$$

Here $q = g/2$, $m_W^2 = q^2\rho^2$, $\mu, \nu = 1, \ldots, 4$, and V_T'' denotes the transverse scalar curvature in the coupled block. Together with the radial Higgs block, the three five-component sectors give a bosonic Hessian of dimension $16L^4 = 65{,}536$ at $L = 8$. The Faddeev–Popov ghost operator is evaluated separately because it enters the functional determinant with a different power.

The run used $a = 2.0 \times 10^{-3}\,\mathrm{GeV}^{-1}$ and $g = 0.65$. Table II summarizes its setup and numerical check. The displayed gauge and ghost spectra are the 16-mode windows for one representative generator; the radial background makes the three gauge blocks identical.

TABLE II. Settings and checks for the September 30, 2026, contained-background $SU(2)$ diagnostic. This run is separate from the scalar production lattice.

Quantity	Setting or result
Lattice	$L = 8$, $a = 2.0 \times 10^{-3}\,\mathrm{GeV}^{-1}$; 4096 sites
Background	Contained Gaussian; reported boundary fraction 1.495×10^{-5}
Gauge	$g = 0.65$, $\xi = 1$, hence $q = 0.325$
Bosonic Hessian	Higgs plus three (A_μ, χ) blocks; $65{,}536 \times 65{,}536$
Spectrum	16 gauge–Goldstone and 16 ghost modes for one generator
Self-adjointness	Bilinear-form residual 2.26×10^{-16}

Figure 9 compares the low spectra with the false-vacuum levels. The first four gauge–Goldstone eigenvalues on the Gaussian background are $20.0715\,\mathrm{GeV}^2$, compared with four near-zero false-vacuum gauge modes; the fifth is $24{,}642.6\,\mathrm{GeV}^2$ versus $25{,}000.0\,\mathrm{GeV}^2$. The lowest ghost eigenvalue is $61.1941\,\mathrm{GeV}^2$, versus a near-zero false-vacuum mode. The four near-zero false-vacuum gauge modes are constant periodic gauge configurations, so a gauge-volume prescription is required before any de-

terminant ratio can be assigned a physical interpretation.

The local counterterm basis and continuum pole residues have been derived and checked for algebraic closure; the reported scalar and background Feynman-gauge pole checks close to 3.11×10^{-15}. Nonlinear scalar-link, Wilson-gauge, and ghost vertex tables have been generated and independently audited against finite transformations and separate action expansions at tested routings. These results validate components of the implementation, not a complete lattice-to-continuum matching calculation.

The latest matching work contains a provisional finite Higgs kinetic-slope estimate, $\delta Z_H = 0.0329117974 \pm 3.60 \times 10^{-7}$, where the quoted uncertainty is QMC statistical error only. A gauge-linked six-point insertion, $\delta\gamma_{\mathrm{gauge}} = -0.01887194 \pm 0.00058491$, and selected higher-point projections are also pilots, not production-precision coefficients. The continuum gauge/ghost two-point tensor, complete finite gauge-sector coefficients, adequate regulator-cancellation and Ward/Slavnov–Taylor closure, and the gauge-volume/collective-coordinate normalization remain unfinished. No physical $SU(2)$ determinant or gauged decay prefactor is reported.

Separate October 14–18 finite-lattice audits addressed translation-mode bookkeeping and trace-estimator stability. A verified-subspace analysis found four-translation tangent capture fractions of 90.8762%, 94.3902%, and 96.1843% in the $L = 24, 30, 36$ spacing samples, respectively. In a distinct paired-probe mode-exchange study, the 256-probe shifts at those sizes agreed with the finite-mode bookkeeping predictions within 0.44 paired standard errors. These analyses use different profiles and probe streams and are not a common spacing extrapolation; they check finite-mode bookkeeping rather than the full trace–log or a continuum limit.

Table III gives the verified-subspace quantities at fixed box extent $Lh = 24$ in the production scaled units. The mode-exchange prediction is $\Delta_{\mathrm{ex}} = \log[\lambda_B/|\lambda_T|]$, where $\lambda_T < 0$ is the fourth translation mode and $\lambda_B > 0$ is the position-selected boundary mode. Figure 10 shows the capture defect against h^2; its dashed guide is descriptive only and is not a continuum extrapolation.

C. $L = 36$ field and collective-mode diagnostics

The separate $L = 36$ run report provides a central line profile and an x–y slice through the relaxed field, with the other coordinates fixed at the lattice midpoint. These are finite-lattice spatial diagnostics and are not merged with the $L = 26$ production determinant results. A separate $L = 36$ low-mode audit recovered the four-dimensional longitudinal translation cluster and one negative longitudinal mode. The transverse orientation mode is counted once in each of the three degenerate transverse channels, giving three internal-orientation modes in the full field. The bar chart in Fig. 11 counts only these identified collective modes, not the full spectrum.

VI. REPRODUCIBILITY AND SOFTWARE

The principal scalar implementation is the single-file program `Higgs_fluctuation_solver_stabilized.py`; its nonlinear vertex library is `su2_lattice_vertices.py`. The program separates deterministic analytic validation from the more expensive research and production stages. Its validation mode compares a matrix-free periodic operator against known Fourier levels and checks complete degeneracies before the bounce calculation. The production scalar stages can be run separately or sequentially, with output summaries and restartable checkpoints. The numerical values in this manuscript are transcribed from the project guide and associated run reports; exact source revision, checkpoint, probe streams, and machine-readable summaries to be archived together during submission. The separate $L = 8$ contained-background gauge–Goldstone and ghost diagnostic is documented in `SU2_Gauge_Coupled_Report.tex`; it is not part of the scalar production or spacing data sets.

VII. DISCUSSION AND CONCLUSIONS

The principal result of this work is a documented numerical chain from a metastable four-component scalar potential through a continuum bounce, finite-lattice stationary background, low-mode classification, stochastic bare trace–log, and continuum-subtracted scalar determinant. Matrix-free operator application makes the large lattice Hessians tractable, while separate channel solves and explicit low-mode deflation preserve the collective-coordinate structure needed by the determinant calculation.

The benchmark does not yet establish a controlled continuum or volume limit. The bare SLQ result is regulator dependent, and the quoted uncertainty on the continuum-subtracted trace–log is not a fully calibrated error estimate. More fundamentally, the ρ^8 leading potential is not closed under its scalar one-loop divergence: the local basis must include ρ^{10} and ρ^{12} operators, whose finite coefficients remain unmatched. The reported continuum determinant is therefore an intermediate subtraction-prescription result rather than a complete matched one-loop EFT prediction. The scalar partial prefactor also retains a missing internal-orientation normalization and requires a physical units convention before it can be interpreted as a rate. These boundaries are part of the result: they separate what the present pipeline computes from what remains a research task.

For the gauge extension, the implemented operators, Ward checks, pole relations, and vertex audits provide prerequisites for a future calculation. The separate $L = 8$ contained-Gaussian test adds low-spectrum checks for the coupled gauge–Goldstone and ghost operators, with a bilinear-form self-adjointness residual of 2.26×10^{-16}. Because its profile is not stationary, this is an operator-validation result, not a decay calculation.

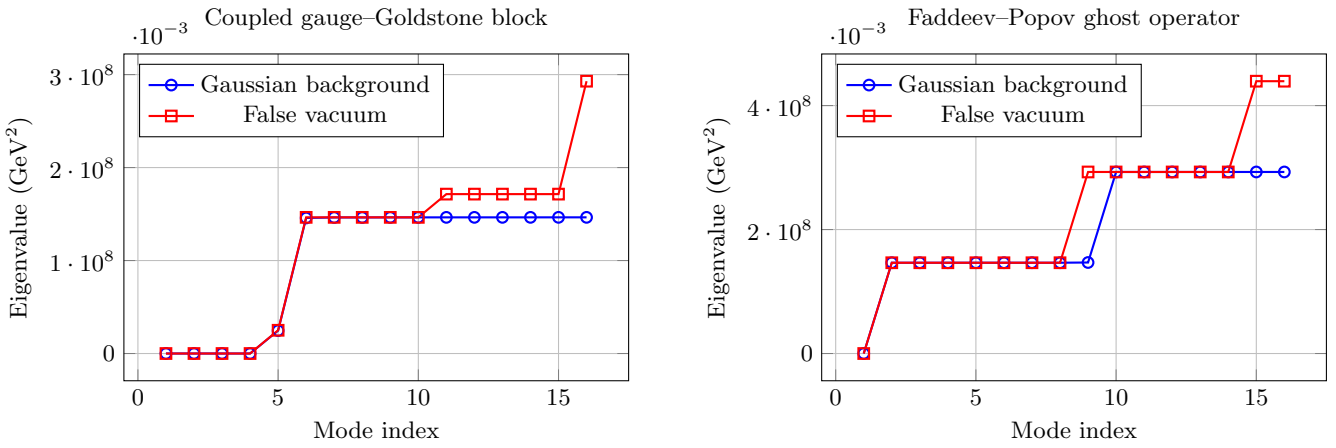


FIG. 9. Low 16 eigenvalues of the coupled gauge–Goldstone block and the Faddeev–Popov ghost operator for one generator in the contained $L = 8$ Gaussian diagnostic. The circles denote the Gaussian-background spectra and the squares the false-vacuum spectra; eigenvalues are in GeV^2. The three gauge–Goldstone blocks are identical. Since the profile is not a stationary bounce, these spectra validate the operator implementation but do not define a gauged decay prefactor.

TABLE III. Verified-subspace spacing diagnostics for the fixed box extent $Lh = 24$ in the production scaled units. C is the four-translation tangent capture fraction, λ_T is the fourth translation eigenvalue, λ_B is the position-selected positive boundary eigenvalue, and Δ_{ex} is the predicted finite-mode exchange. These data audit mode bookkeeping and do not constitute a full-determinant spacing extrapolation.

L	h	C (%)	$(1-C)/h^2$	λ_T	λ_B	Δ_{ex}
24	1	90.8762	0.09124	-3.23612×10^{-3}	0.268131	4.41710
30	0.8	94.3902	0.08765	-2.12606×10^{-3}	0.273198	4.85592
36	2/3	96.1843	0.08585	-2.46598×10^{-4}	0.281746	7.04100

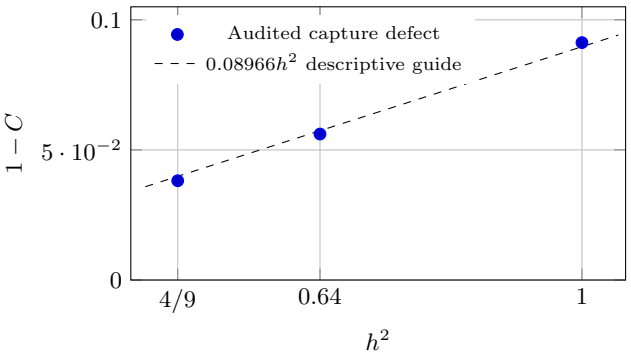


FIG. 10. Four-translation tangent-capture defect from the verified-subspace spacing audit at fixed $Lh = 24$. The dashed line is the through-origin descriptive guide $1 - C = 0.08966h^2$ from the three points; it has no inferential uncertainty and is not a controlled continuum extrapolation.

They do not replace finite matching, a completed coupled gauge–Goldstone– ghost determinant, or tests of regulator, gauge-parameter, spacing, and volume independence. Any later comprehensive study should report those quantities only after the corresponding matching coefficients and determinant have been completed and independently validated.

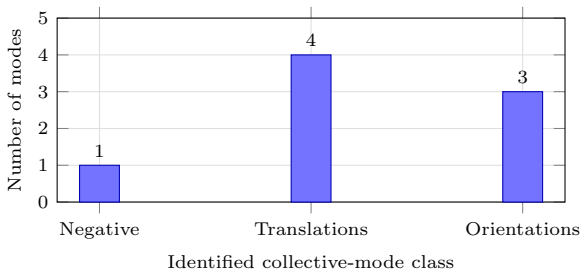


FIG. 11. Identified $L = 36$ collective-mode counts: one negative mode, four translations, and three internal orientations. Other eigenpairs are omitted.

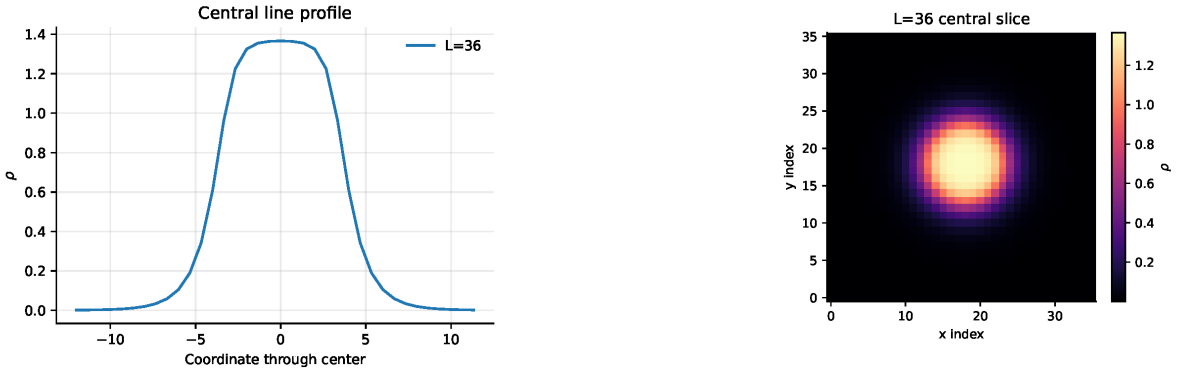


FIG. 12. $L = 36$ relaxed-field diagnostics: central line profile (left) and central x–y slice (right). These show finite-lattice geometry, not a continuum extrapolation or an additional determinant result.

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Data availability. The source and outputs are not yet archived; during submission, we will deposit the code, dependencies, run metadata, probes, and result tables and add the repository identifier.

[1] S. Coleman, "The Fate of the False Vacuum: Semiclassical Theory," Phys. Rev. D **15**, 2929 (1977); Erratum, Phys. Rev. D **16**, 1248 (1977).
[2] C. G. Callan, Jr. and S. Coleman, "The Fate of the False Vacuum. II. First Quantum Corrections," Phys. Rev. D **16**, 1762 (1977).
[3] J. R. Espinosa, "Vacuum Decay in the Standard Model: Analytical Results with Running and Gravity," J. Cosmol. Astropart. Phys. **06**, 052 (2020).
[4] V. Guada and M. Nemevšek, "Exact one-loop false vacuum decay rate," Phys. Rev. D **102**, 125017 (2020).
[5] W.-Y. Ai, J. S. Cruz, B. Garbrecht, and C. Tamarit, "Gradient effects on false vacuum decay in gauge theory," Phys. Rev. D **102**, 085001 (2020).
[6] W.-Y. Ai, J. Alexandre, and S. Sarkar, "False vacuum decay rates, more precisely," Phys. Rev. D **109**, 045010 (2024).
[7] P. Baratella, M. Nemevšek, Y. Shoji, K. Trailović, and L. Ubaldi, "Functional determinants for false vacuum decay," J. High Energy Phys. **05**, 100 (2025).
[8] P. Baratella, M. Nemevšek, Y. Shoji, K. Trailović, and L. Ubaldi, "Revising the full one-loop gauge prefactor in electroweak vacuum stability," Phys. Rev. Lett. **134**, 011601 (2025).
[9] G. Laverda and J. Rubio, "The rise and fall of the Standard-Model Higgs: electroweak vacuum stability during kination," J. High Energy Phys. **05**, 339 (2024).
[10] J. Shen, P. Draper, and A. X. El-Khadra, "Vacuum decay and Euclidean lattice Monte Carlo," Phys. Rev. D **107**, 094506 (2023).
[11] L. Jin and J. Swaim, "Non-perturbative False Vacuum Decay Using Lattice Monte Carlo in Imaginary Time," arXiv:2512.00542 (2025; revised 2026).
[12] S. Coleman, V. Glaser, and A. Martin, "Action Minima among Solutions to a Class of Euclidean Scalar Field Equations," Commun. Math. Phys. **58**, 211 (1978).
[13] I. M. Gel'fand and A. M. Yaglom, "Integration in Functional Spaces and Its Applications in Quantum Physics," J. Math. Phys. **1**, 48 (1960).
[14] A. J. McKane and M. B. Tarlie, "Regularisation of Functional Determinants Using Boundary Perturbations," J. Phys. A **28**, 6931 (1995).
[15] J. Baacke and G. Lavrelashvili, "One-Loop Corrections to the Metastable Vacuum Decay," Phys. Rev. D **69**, 025009 (2004).
[16] G. V. Dunne and K. Kirsten, "Functional Determinants for Radial Operators," J. Phys. A **39**, 11915 (2006).
[17] K. Kirsten and A. J. McKane, "Functional determinants for general Sturm–Liouville problems," J. Phys. A **37**, 4649 (2004).
[18] G. V. Dunne, "Functional determinants in quantum field theory," J. Phys. A: Math. Theor. **41**, 304006 (2008).
[19] D. V. Vassilevich, "Heat kernel expansion: User's manual," Phys. Rep. **388**, 279 (2003).
[20] M. F. Hutchinson, "A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines," Commun. Stat. Simul. Comput. **18**, 1059 (1989).
[21] S. Ubaru, J. Chen, and Y. Saad, "Fast Estimation of tr($f(A)$) via Stochastic Lanczos Quadrature," SIAM J. Matrix Anal. Appl. **38**, 1075 (2017).
[22] U. Banerjee, J. Chakrabortty, and K. Ramkumar, "Renormalization of scalar and fermion interacting field theory for arbitrary loop: Heat-Kernel approach," Eur. Phys. J. Plus **139**, 714 (2024).
[23] K. G. Wilson, "Confinement of quarks," Phys. Rev. D **10**, 2445 (1974). doi:10.1103/PhysRevD.10.2445.